Experiment-6

Write a Servlet to demonstrate session tracking using HttpSession.

AIM:

To demonstrate session tracking using **HttpSession** in a Servlet-based web application. The exercise involves implementing a simple login system where a user can log in with their username and password, and their session will be tracked throughout their interaction with the web application.

ALGORITHM:

1. Input:

- The user enters their username and password in a login form.
- The servlet validates the credentials and creates an HTTP session for the user if authentication is successful.

2. Steps for Implementing Session Tracking with HttpSession:

Step 1: Create the Login Form (HTML)

- Create an HTML form with two input fields:
  - Username
  - Password
- Provide a submit button to send the data to the servlet for authentication.

Step 2: Create the Login Servlet

- Write a Java servlet class to handle login requests.
- The servlet will:
  - Retrieve the username and password from the form.
  - Authenticate the user by comparing the credentials to a stored value (e.g., hardcoded values or from a database).
  - Create a new HTTP session for the user if the credentials are valid.
  - Set session attributes such as the user's username to track the session.

Step 3: Validate User Credentials

- In the servlet, check whether the entered username and password match the expected values.
- If the credentials are valid, create a session and store the username in the session.
- If the credentials are invalid, redirect the user back to the login page with an error message.

Step 4: Create and Track User Session

- Upon successful login, use the **HttpSession** object to store the session data.
- Use the method:
  `session.setAttribute("username", username)` to save the username.
- Optionally, set session timeout using:
  `session.setMaxInactiveInterval(time_in_seconds)` to manage session expiration.

Step 5: Redirect to a Protected Page

- After a successful login, redirect the user to a protected page (e.g., dashboard or home page).
- The protected page should verify whether the user is authenticated by checking the session attribute.

Step 6: Logout and Session Invalidation

- Provide a logout feature that invalidates the session using: `session.invalidate()` when the user logs out.
- After logout, redirect the user back to the login page.

Step 7: Session Expiration

- If the session expires or the user manually logs out, the session will be invalidated.
- The user will then be redirected to the login page.

CODE:

Login.html:

```
<!DOCTYPE html>
<html>
<head>
    <title>Login - School Management</title>
    <style>
        /* ===== Body & Background ===== */
        body {
            margin: 0;
            height: 100vh;
            background-image: url('School image2.png'); /* optional school image */
            background-size: 50%;
            background-repeat: no-repeat;
            background-position: center;
            background-attachment: fixed;
            display: flex;
            justify-content: center;
            align-items: center;
            font-family: Arial, Helvetica, sans-serif;
        }

        /* ===== Login Box ===== */
        .login-container {
            background: rgba(0,0,0,0.75);  /* dark overlay */
            padding: 40px 50px;
            border-radius: 12px;
            box-shadow: 0 8px 25px rgba(0,0,0,0.5);
            width: 350px;
            color: #fff;
        }

        .login-container h2 {
```

```
        text-align: center;
        margin-bottom: 25px;
        font-size: 28px;
        color: #4CAF50;
    }

    /* ===== Input Fields ===== */
    .login-container input[type="text"],
    .login-container input[type="password"] {
        width: 100%;
        padding: 12px;
        margin: 10px 0;
        border-radius: 6px;
        border: none;
        font-size: 14px;
    }

    /* ===== Buttons ===== */
    .login-container button {
        width: 100%;
        padding: 12px;
        background-color: #4CAF50;
        color: white;
        border: none;
        border-radius: 6px;
        font-size: 16px;
        cursor: pointer;
        font-weight: bold;
        margin-top: 10px;
    }

    .login-container button:hover {
        background-color: #45a049;
    }

    /* ===== Register Link ===== */
    .register-link {
        text-align: center;
        margin-top: 15px;
        font-size: 14px;
    }

    .register-link a {
        color: #4CAF50;
        text-decoration: none;
        font-weight: bold;
    }

    .register-link a:hover {
        text-decoration: underline;
    }

    /* ===== Responsive ===== */
    @media (max-width: 400px) {
        .login-container {
```

```
                width: 90%;
                padding: 30px;
            }
        }
    </style>
</head>
<body>

<div class="login-container">
    <h2>Login</h2>
    <form action="login" method="post">
        <input type="text" name="username" placeholder="Username" required>
        <input type="password" name="password" placeholder="Password" required>
        <button type="submit">Login</button>
    </form>
    <div class="register-link">
        <p>Don't have an account? <a href="Register.html">Register here</a></p>
    </div>
</div>

</body>
</html>
```

LoginServlet.java:

```
package WTMA;

import java.io.IOException;
import java.io.PrintWriter;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.*;

@WebServlet("/login")
public class LoginServlet extends HttpServlet {

    private static final long serialVersionUID = 1L;

    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        String uname = request.getParameter("username");
        String pass = request.getParameter("password");

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();


        if ("admin".equals(uname) && "1234".equals(pass)) {


            HttpSession session = request.getSession();

            session.setAttribute("user", uname);
```

```
            response.sendRedirect("welcome");

        } else {
            out.println("<h3>Invalid Login</h3>");
            out.println("<a href='Login.html'>Try Again</a>");
        }
    }
}
```

LogoutServlet.java:

```
package WTMA;
import java.io.IOException;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.*;
@WebServlet("/logout")
public class LogoutServlet extends HttpServlet {
   private static final long serialVersionUID = 1L;

   protected void doGet(HttpServletRequest request, HttpServletResponse response)

        throws ServletException, IOException {

      HttpSession session = request.getSession(false);

      if (session != null) {

         session.invalidate();

      }

      response.sendRedirect("Login.html");

   }

}
```

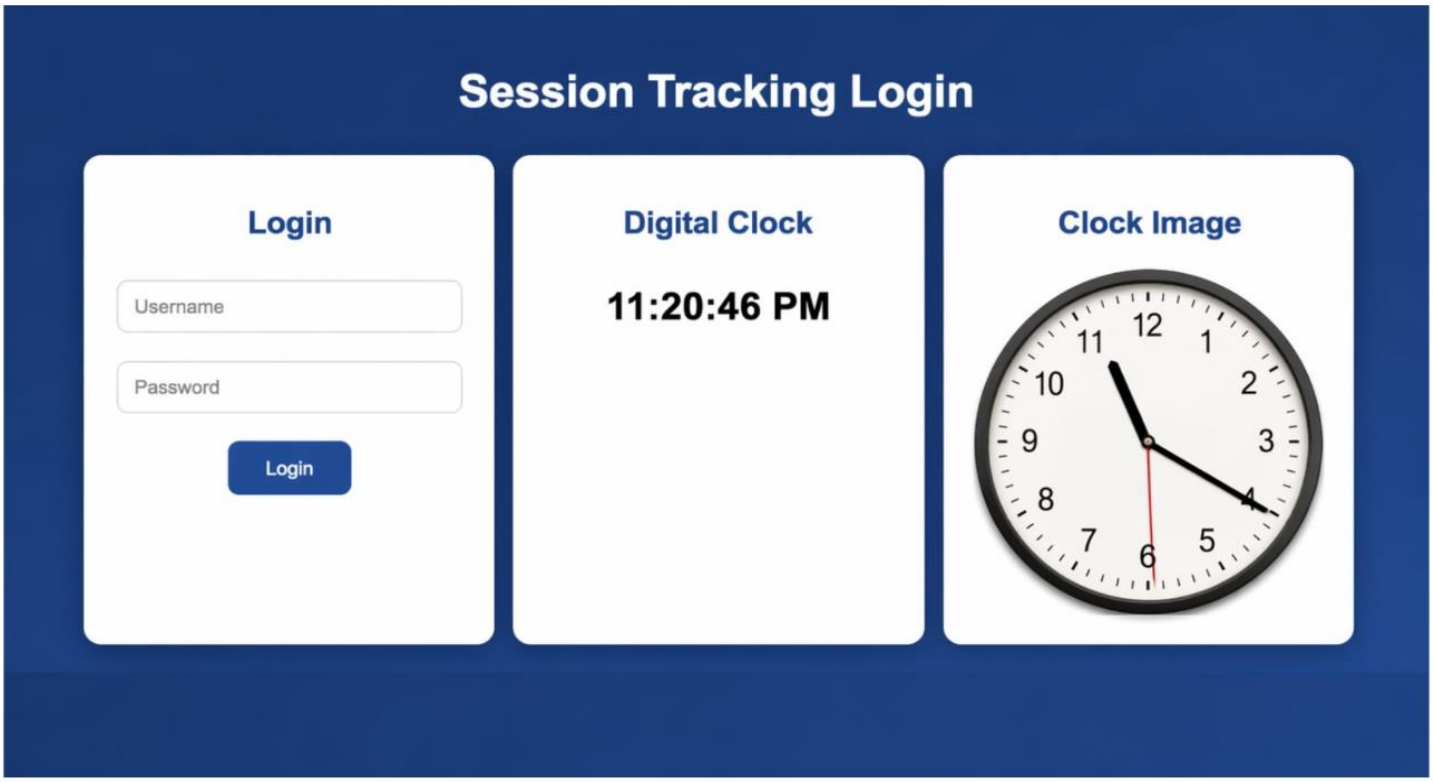
WelcomeServlet.java:

```
package WTMA;

import java.io.IOException;

import java.io.PrintWriter;
```

```
import jakarta.servlet.ServletException;

import jakarta.servlet.annotation.WebServlet;

import jakarta.servlet.http.*;

@WebServlet("/welcome")

public class WelcomeServlet extends HttpServlet {

    protected void doGet(HttpServletRequest request, HttpServletResponse response)

            throws ServletException, IOException {

        response.setContentType("text/html");

        PrintWriter out = response.getWriter();

        HttpSession session = request.getSession(false);

        if (session != null && session.getAttribute("user") != null) {

            String user = (String) session.getAttribute("user");

            out.println("<!DOCTYPE html>");

            out.println("<html>");

            out.println("<head>");

            out.println("<meta charset='UTF-8'>");

            out.println("<title>Dashboard</title>");

            out.println("<style>");

            out.println("body { margin:0; font-family: Arial; background:#eef2f7; }");

            out.println(".header { background:#1e3d59; color:white; padding:20px; text-align:center; font-size:26px; }");

            out.println(".nav { background:#3c6382; padding:10px; text-align:center; }");

            out.println(".nav a { color:white; margin:10px; text-decoration:none; font-weight:bold; }");

            out.println(".box { width:50%; margin:50px auto; background:white; padding:25px; border-radius:10px; box-shadow:0
0 10px #aaa; text-align:center; }");

            out.println(".btn { padding:10px 20px; background:red; color:white; text-decoration:none; border-radius:5px; }");
```

```
        out.println("</style>");
        out.println("</head>");
        out.println("<body>");
        out.println("<div class='header'>School Management System</div>");
        out.println("<div class='nav'>");
        out.println("<a href='#'>Home</a>");
        out.println("<a href='#'>Students</a>");
        out.println("<a href='#'>Teachers</a>");
        out.println("<a href='#'>Courses</a>");
        out.println("</div>");
        out.println("<div class='box'>");
        out.println("<h2>Welcome, " + user + " </h2>");
        out.println("<p>You have successfully logged in.</p>");
        out.println("<a class='btn' href='logout'>Logout</a>");
        out.println("</div>");
        out.println("</body>");
        out.println("</html>");

      } else {
        response.sendRedirect("Login.html");
      }
    }
}
```

Output:


Session Tracking Login
Login
Username
Password
Login
Digital Clock
11:20:46 PM
Clock Image

Result:

The session tracking mechanism using **HttpSession** in a Servlet-based web application was successfully implemented. The user was able to log in using valid credentials, and a session was created to store user-specific data.